

SLOWLAB: Separating Experimental Allocation, Evidence Use, and Recommendation under Slow and Irreversible Feedback

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Abstract

Scientific-agent benchmarks rarely test whether an agent can allocate fixed physical capacity, interpret evidence that arrives during long-running trials, and commit to an operating recommendation when the experiments themselves incur economic loss. We introduce SLOWLAB, a controlled-environment benchmark whose core suite covers interface calibration, split-plot optimisation, and recommendation after an energy-price shift, with immutable 210-day trials, correlated noise, and separate design-policy and history-reader roles. Exact hidden-site regret supports model comparisons, while a common-history Bayes-risk diagnostic asks what an executed design made knowable and how much excess risk remains in the agent’s recommendation. Across 928 frozen model episodes and a separately frozen 384-episode noise extension, standard design and inference aids do not reliably reduce final regret for the tested Luna and Sol conditions, and a prespecified half-noise analysis finds a harmful design-aid effect ($\Delta = 0.0113$, 95% CI $[0.0053, 0.0184]$, adjusted $p = 0.0096$).

1 Introduction

Agents that operate scientific experiments must do more than answer questions about a finished table. They choose what to run, allocate scarce physical capacity, interpret partial and noisy evidence, and decide when that evidence supports an operating recommendation. Existing work evaluates important parts of this loop: BoxingGym limits sequential experimentation and measures information gain, Olympus and Summit expose executable optimisation surfaces, Minerva studies noisy parallel batches, and budgeted or delayed-feedback Bayesian optimisation formalises cost and latency [1–3, 5, 8, 9]. The combined decision remains difficult to measure when trials occupy a tiered facility, poor experiments lose real objective value, and a submitted treatment cannot be recalled.

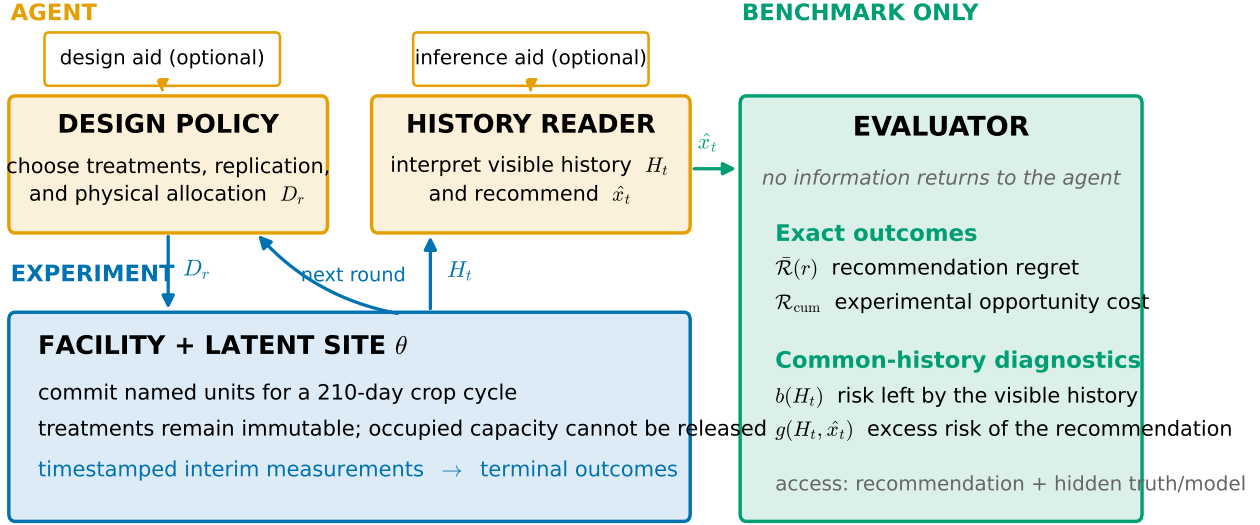
SLOWLAB makes that decision executable in controlled-environment horticulture. A site is drawn from a documented crop-and-economics distribution. An agent assigns treatments to named chambers and hydroponic loops, commits them for a 210-day crop cycle, observes only measurements available by the current simulation time, and recommends an operating point. Chamber-level climate controls and loop-level cultural controls create a split-plot action space: replication, treatment coverage, and physical placement compete for the same units. The published TOMGRO crop model supplies traceability rather than author-independent difficulty; budgets, ranges, observation channels, and site distributions remain explicit benchmark choices [6].

We organise the benchmark around three roles (Figure 1). A *design policy* chooses treatments and their physical allocation. A *history reader* maps the time-stamped visible record to a recommendation. A closed-loop agent may implement both in one model, but fixed-history replay can vary one role while holding the other fixed. The *evaluator* belongs to the benchmark, may use hidden truth to score the submitted outputs, and never returns privileged information to the agent.

Our contribution is threefold. First, we formulate scientific-agent evaluation as coupled allocation, evidence-use, and recommendation problems under slow, noisy, costly, and irreversible feedback. Second, we define an information boundary and a common-history diagnostic that separates what the executed design could support from how the reader used that evidence. Third, we test scripted policies and model agents in frozen, site-paired experiments. The resulting claim is deliberately bounded: the tested aids show no reliable improvement in the primary setting, one lower-noise condition shows harm, and the relative importance of allocation and reading varies by task.

2 Decision problem and roles

An instance is a latent site $\theta \sim P_\Theta$ defining an objective $f_\theta : \mathcal{X} \rightarrow \mathbb{R}$. In round r , the agent observes a timestamped history H_r , chooses a finite design D_r , and reports a recommendation $\hat{x}_r$. A design contains factor settings and an assignment to available physical units $\mathcal{U}_r$; feasibility therefore depends on the facility, not only on whether $x \in \mathcal{X}$.



A closed-loop agent may implement design and reading in one model; fixed-history replay separates the roles experimentally.

Figure 1: **Roles and information flow.** The design policy commits treatments and physical units; the history reader sees only the time-stamped history and produces a recommendation. The benchmark-side evaluator computes exact outcomes from hidden truth and calibrated diagnostics from the same visible history. Interim observations may update the reader and the next design, but cannot alter a committed treatment or release capacity.

The *design policy* π_D maps visible history and remaining resources to a design,

$$D_r = \pi_D(H_r, \mathcal{U}_r) \in \mathcal{D}. \quad (1)$$

It controls what evidence can exist. The *history reader* ρ maps the same visible history to beliefs and a recommendation,

$$\hat{x}_r = \rho(H_r) \in \mathcal{X}. \quad (2)$$

Equations 1 and 2 make the separation explicit: the first role controls what evidence can exist, while the second controls how acquired evidence is used. These are analytical roles rather than required software modules; the same LLM produces both outputs in a closed-loop episode.

Submitting D_r occupies its units for a fixed duration Δ . Measurements enter H_t only when their timestamps are reached. The benchmark permits inspection during a cycle but does not permit treatment modification, early release, or refunds. Terminal outcomes follow

$$y_u = f_\theta(x_u; \phi_u) + \varepsilon_u, \quad \varepsilon \sim \mathcal{N}(0, \Sigma(D_r)), \quad (3)$$

Equation 3 makes the observation hierarchy explicit: per-unit biological variation ϕ_u and chamber, loop, batch, and measurement noise make the covariance structured and treatment dependent. A poor design can therefore fail by choosing uninformative treatments, confounding them with blocks, or allocating too little replication.

The *evaluator* is outside the episode policy. It receives designs and recommendations and may access θ , f_θ , P_Θ , and the forward model. These objects never enter H_t , and evaluator outputs are not fed back to the agent. A design aid may change π_D ; an inference aid may change ρ ; neither is an evaluator.

The four regime properties follow directly. Feedback is *slow* because finite parallel capacity and latency bound evidence per calendar time. It is *noisy* because one unit does not settle a comparison. It is *costly* because experiments earn the same economic objective they seek to optimise. It is *irreversible* because committed capacity and treatments cannot be recovered after later evidence arrives.

Task	factors	rounds	units/round	unit-days	decision tested
Sanity	1	2	8	3,360	interface and calibration
Optimise	2	3	12	7,560	split-plot iteration
Transfer	4	3	24	15,120	post-shock recommendation

Table 1: Core Version 2 tasks. Unit-days are committed physical capacity.

3 Benchmark instantiation

3.1 World and facility

A site seed fixes crop parameters, climate, input prices, and output prices. The forward model combines reduced TOMGRO physiology with an explicit greenhouse gross-margin model. Climate chambers are whole plots and independently plumbed hydroponic loops are subplots. Temperature, photoperiod, lighting, and carbon dioxide are chamber-level controls; density and pruning are loop-level controls. Observations retain plant, loop, chamber, batch, measurement, and reporting-resolution variation at their actual scopes.

Each submission names treatments, assigns every treatment to physical units, and supplies a randomisation seed. Mechanical validation is free and checks capacity, factor bounds, and control hierarchy. It does not certify statistical quality. Rejected submissions consume no experimental budget but remain recorded as agent failures.

3.2 Core tasks

Table 1 summarises the Version 2 core suite. Its three tasks share the same objective and commitment rule, but differ in the decision being tested rather than serving as nominal difficulty levels.

SANITY is a one-factor closed-loop and evaluator check. OPTIMISE varies one chamber-level and one loop-level factor, making allocation and block structure unavoidable. TRANSFER varies four resource factors; after the campaign, electricity and gas prices rise by $2.2\times$ and the agent must recommend again without new trials.

The implementation retains an experimental six-factor SCREEN configuration, but we exclude it from the core suite and empirical claims. Its present endpoint is operating regret, which measures scarce high-dimensional optimisation rather than the stated screening construct. A valid future version should elicit factor ranks, signs, and uncertainty and score them directly.

3.3 Observation timing

The agent may advance simulation time and request registered modalities from selected units. Only records whose timestamps have passed are returned. Inspection can update the current recommendation and, before the next round, its next design. In the synchronous action space it cannot alter the running treatment, release a unit, or create a replacement batch.

4 Evaluation

4.1 Exact outcome evaluator

The primary evaluator uses the hidden site only after an action is submitted. Final simple regret is

$$\mathcal{R}_{\text{final}}(\theta, \hat{x}) = f_{\theta}(x_{\theta}^*) - f_{\theta}(\hat{x}), \quad x_{\theta}^* \in \arg \max_{x \in \mathcal{X}} f_{\theta}(x). \quad (4)$$

Experimental opportunity cost charges every occupied unit for the objective it forwent,

$$\mathcal{R}_{\text{cum}} = \Delta \sum_{r=1}^R \sum_{u \in \mathcal{U}_r} [f_{\theta}(x_{\theta}^*) - f_{\theta}(x_u)]. \quad (5)$$

Equations 4 and 5 are exact up to numerical optimisation of the hidden response surface and support the model and tool comparisons. They are reported separately: an agent may find a good recommendation through an economically poor campaign.

4.2 Common-history diagnostic

Outcome regret alone cannot say whether a failure came from missing evidence or poor use of available evidence. For the history visible at time t , define remaining Bayes risk

$$b(H_t) = \min_{x \in \mathcal{X}} \mathbb{E}[\mathcal{R}_{\text{final}}(\theta, x) \mid H_t] \quad (6)$$

and recommendation excess risk

$$g(H_t, \hat{x}_t) = \mathbb{E}[\mathcal{R}_{\text{final}}(\theta, \hat{x}_t) \mid H_t] - b(H_t) \geq 0. \quad (7)$$

In Equations 6 and 7, large $b(H_t)$ indicates that the visible history still leaves a difficult decision, while large $g(H_t, \hat{x}_t)$ indicates that the recommendation is worse than the best action supported by that same history. Both condition on identical information and therefore avoid the Version 1 comparison between a preposterior design score and a realised recommendation.

The diagnostic evaluator uses independent posterior samples for action selection and evaluation. Crop and economic factors update separately until terminal revenue couples them. We report diagnostics only where calibration error is smaller than the interpreted effect; all primary claims remain based on exact hidden-site outcomes.

5 Experimental design

5.1 Fixed-history role separation

We first create completed histories using random spread, split-plot DOE, and constraint-aware batch Bayesian optimisation. Ordinary, block-aware, and component-aware GP readers receive copies of each accepted history and produce recommendations without changing any design or observation. Crossing design source with reader makes row differences allocation effects, column differences evidence-use effects, and residuals interactions. Privileged readers are labelled as diagnostics and never treated as deployable baselines.

5.2 Frozen closed-loop study

The primary matrix runs Luna on OPTIMISE with bare, design-aid, inference-aid, and combined interfaces: 56 sites, two provider generation seeds per site, and 448 episodes. Prespecified secondary matrices add a 24-site Sol bare-versus-inference replication, Luna task heterogeneity on SANITY and TRANSFER, within-cycle versus terminal-only access, and a prompt variant, for 928 frozen episodes. A separately frozen adaptive extension reruns the four Luna interfaces on 16 fresh sites at 0.5 $\times$, 1 $\times$, and 2 $\times$ observation noise, adding 384 episodes.

Sites are statistical units; provider generations are within-site replicates. We average generation seeds within site before forming paired differences and use 10,000 site-bootstrap resamples for percentile 95% intervals. The primary tool family uses Holm correction; prespecified secondary families use Benjamini–Hochberg correction. Transport failures may be retried with the same identity. Format failures, infeasible submissions, and model early stopping remain outcomes.

5.3 Post-primary model extension

A frozen prospective extension compares Luna, DeepSeek V4 Flash 0731, and Qwen 3.8 27B on the same 24 fresh OPTIMISE sites. Each model runs bare and inference-aid conditions with two provider generation seeds, for 288 episodes. This extension was specified after the original Phase 4 analysis and will be reported as post-primary evidence. It estimates three within-model tool contrasts and same-site pairwise model contrasts; it does not alter the original primary family.

6 Results

6.1 Allocation and reading are separable and task dependent

Across held-out fixed histories, the overall range across scripted design sources is 0.01718 regret, the range across deployable readers is 0.00619, and the design–reader interaction RMS is 0.00209. This aggregate does not define a universal ordering. The design range is largest on TRANSFER (0.04727), whereas on SANITY the reader range

Policy	Reader / aid	Sites $\times$ repeats	Mean regret	Δ vs. bare [95% CI]	adjusted p
Random spread	ordinary GP	8×1	0.0072	descriptive	—
Split-plot DOE	ordinary GP	8×1	0.0128	descriptive	—
Batch BO	ordinary GP	8×1	0.0066	descriptive	—
Luna	none	56×2	0.0128	reference	—
Luna	design	56×2	0.0129	+0.0002 [-0.0030, +0.0032]	1.000
Luna	inference	56×2	0.0108	-0.0020 [-0.0052, +0.0013]	0.727
Luna	both	56×2	0.0126	-0.0002 [-0.0037, +0.0033]	1.000
Sol	none	24×2	0.0030	reference	—
Sol	inference	24×2	0.0035	+0.0005 [-0.0006, +0.0017]	0.381

Table 2: Final simple regret on OPTIMISE (lower is better). Scripted rows are fixed-history reader replays on eight Phase 3 sites and are descriptive; they are included as reference points, not tested against the independently sampled model runs. Phase 4 model contrasts average two provider generations within each site before inference. Luna tool contrasts use Holm correction across the three preregistered comparisons; the Sol replication uses the prespecified secondary-family adjustment.

Contrast	sites	Δ regret	95% CI	adjusted p
Luna, T1: inference – bare	24	-0.0008	[-0.0053, +0.0037]	0.722 (BH)
Luna, T4: inference – bare	24	+0.0114	[-0.0017, +0.0287]	0.313 (BH)
Luna, within-cycle – terminal-only	24	+0.0077	[-0.0004, +0.0166]	0.085 (BH)
Luna, 0.5 $\times$ noise: design – bare	16	+0.0113	[+0.0053, +0.0184]	0.010 (BH)
Luna, 2 $\times$ noise: design – bare	16	+0.0048	[-0.0047, +0.0131]	0.605 (Holm)
Luna, 2 $\times$ noise: inference – bare	16	-0.0067	[-0.0155, +0.0012]	0.374 (Holm)
Luna, 2 $\times$ noise: both – bare	16	-0.0040	[-0.0116, +0.0032]	0.605 (Holm)

Table 3: Held-out secondary and robustness contrasts (positive values are worse). The half-noise design-aid contrast is the only listed interval that excludes zero.

exceeds the design range. A block-aware reader also fails to improve over the ordinary GP on its independent calibration set (paired normalised-regret improvement -0.0112 , SE 0.0095 , 24 pairs). Structural readers therefore serve as diagnostics rather than assumed stronger baselines. Table 2 places these fixed-history reference points beside the closed-loop results without testing across independently sampled sites.

6.2 Standard aids do not reliably reduce final regret

For Luna on the 56 primary sites, the design-aid difference is $+0.00016$ (95% CI $[-0.00299, 0.00321]$), inference aid is -0.00196 ($[-0.00520, 0.00133]$), and both aids are -0.00023 ($[-0.00365, 0.00326]$). None is significant after Holm correction. Sol has lower descriptive regret on its independent sites, but its inference aid also fails to improve its bare arm: $+0.00053$ ($[-0.00055, 0.00173]$), adjusted $p = 0.381$. Because the original Luna and Sol matrices use different sites, their absolute levels are descriptive, not a paired model ranking.

Table 3 reports the prespecified secondary and noise-robustness families. The aids were used: inference recommendations were adopted frequently, while geometric design adoption was lower. The null effects therefore do not reduce to complete tool avoidance. They show that the tested delivery format did not reliably convert adoption into lower endpoint regret.

Within-cycle access changes model behaviour and API use, but its paired final-regret difference is $+0.00768$ (95% CI $[-0.00041, 0.01662]$, adjusted $p = 0.085$). Under the synchronous action space, interim evidence can update a recommendation but has no guaranteed path to modify running capacity. Testing its operational value requires early termination, staggered release, or treatment modification.

6.3 Noise robustness and transfer

At twice the registered observation noise, all three Luna tool-minus-bare intervals cross zero. At half noise, the design aid increases regret by 0.01133 (95% CI $[0.00530, 0.01842]$, adjusted $p = 0.0096$). Other half-noise and one-noise tool contrasts, and the registered noise interactions, are not significant. The supported conclusion is therefore

Condition	sites $\times$ repeats	pre-shock	post-shock
Luna bare	24×2	0.0602 ± 0.0054	0.0681 ± 0.0095
Luna inference	24×2	0.0716 ± 0.0074	0.1133 ± 0.0140
Inference – bare (post-shock)	24 sites	$+0.0452 [+0.0200, +0.0781]$	

Table 4: TRANSFER regret before and after the held-out $2.2\times$ energy-price shock (mean $\pm$ site-level SE). The post-shock paired interval is descriptive because this contrast was not a preregistered inferential family.

bounded: the tested aids show no reliable benefit, and one lower-noise condition shows harm.

Table 4 separates pre-shock and post-shock regret. After the held-out $2.2\times$ energy-price shock, the inference-aid arm has higher regret than bare by 0.0452, with a descriptive paired site-bootstrap interval $[0.0200, 0.0781]$. This contrast was not a registered inferential family and motivates a targeted replication rather than a confirmatory tool claim.

6.4 Evaluator scope

On SANITY, independent 32×32 crop–economics product posteriors agree to response-curve RMSE 0.00445 and select nearly identical Bayes actions. This check does not establish equal error in higher dimensions. We therefore use exact realised regret for the claims above and restrict $b(H_t)$ and $g(H_t, \hat{x}_t)$ to calibrated mechanism analyses.

7 Related work

BoxingGym gives agents a finite sequence of experimental queries and evaluates prediction and expected information gain; it also reports that additional observations do not help uniformly [3]. SLOWLAB instead makes physical allocation and experimental opportunity cost part of the action and separates executed-design support from recommendation quality. CARE evaluates LLM-guided reaction optimisation under a fixed condition budget; our emphasis is correlated split-plot allocation and evidence use in a closed-loop physical campaign [7].

Olympus and Summit provide executable benchmarks for experimental optimisation, while Minerva adds noisy parallel batches and shared-resource constraints [2, 5, 8]. These systems motivate realistic surfaces and planners. SLOWLAB focuses on the information boundary between an agent that operates the campaign and a benchmark-side evaluator that knows the simulator.

Budgeted Bayesian optimisation studies finite or unknown evaluation costs, and delayed feedback work studies pending evaluations [1, 9]. Our costs are realised in the objective itself, while tiered controls turn unit assignment into a split-plot design problem [4]. The present synchronous action space exposes within-cycle measurements but does not yet include the staggered replacement actions needed to realise their full value.

8 Limitations

The validated domain is simulated greenhouse tomato production. Published physiology and external price sources improve traceability, but parameter distributions, facility size, factor ranges, and observation channels remain benchmark choices. Results do not establish the same tool effects in chemistry, materials, biology, field agriculture, or a physical greenhouse.

The original frozen model study contains Luna and a smaller Sol replication through one provider interface. A post-primary same-site Luna–DeepSeek–Qwen extension improves model coverage but cannot retroactively enlarge the original confirmatory family. Provider routing, model updates, and prompt format remain sources of external variation.

Within-cycle observation is only consequential through the current recommendation and the next synchronous design. The environment lacks treatment modification, early stopping, cleanup cost, residual value, and staggered replacement. It therefore tests whether agents use interim evidence, not the operational value of adaptive physical intervention.

The common-history evaluator is well calibrated only on the one-factor task. Its finite candidate search is a lower bound on the world optimum, and its higher-dimensional error is not fully characterised. Exact hidden-site regret consequently carries the primary claims.

Finally, the experimental SCREEN configuration does not yet elicit or score factor selection. Excluding it from the core suite avoids presenting high-dimensional operating regret as evidence of screening ability; a future task needs direct rank, sign, and uncertainty targets.

9 Conclusion

SLOWLAB evaluates an agent as the operator of a resource-limited experimental campaign. The design policy determines what evidence can exist, the history reader determines how that evidence becomes a recommendation, and the evaluator measures both without leaking hidden truth into the episode. This separation turns a vague failure to “reason scientifically” into testable allocation, evidence-use, and recommendation questions.

The current experiments support a narrower conclusion than the Version 1 mechanism story. Allocation and reading are empirically separable but task dependent. Standard design and inference aids do not reliably improve endpoint regret in the tested primary conditions; a fresh high-noise extension preserves that null result, and one lower-noise condition shows harm. Within-cycle access changes behaviour without improving the final recommendation in the current synchronous action space. The next decisive tests are matched cross-model comparisons and action spaces in which interim evidence can change physical capacity.

References

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A Protocol and reproducibility

The frozen experiment families are:

- Primary: `slowlab-v2-phase4-confirmatory-2026-09-13-a2`, with 928 complete and unique expected episode identities.

- Noise extension: `slowlab-v2-phase4-noise-robustness-2026-09-19-a2`, with 384 complete rows.
- Post-primary model extension: `slowlab-v2-phase5-model-extension-2026-09-20-a1`.

Each record stores requested and actual model identifiers, provider, generation seed, prompt hashes, usage, cost, retry attempt, finish reason, format failures, infeasible submissions, and the accepted event history.

The public numerical source for the current tables is `results/phase5_paper_summary_env2.0.0.json`. From an unpacked frozen data bundle, `scripts/reproduce_phase5_paper.py` rebuilds analyses, generated tables, figures, claim checks, cross-reference checks, and the PDF. Private review notes are excluded from the repository artifact.

B Reference policies

Random spread maximises geometric coverage subject to the same facility validator as model agents. Split-plot DOE balances whole-plot combinations across chambers and randomises loop-level settings within chamber. Constraint-aware batch Bayesian optimisation fits an ordinary GP to completed outcomes and proposes a feasible batch by sequential fantasisation. Reader replay copies each accepted completed history to ordinary, block-aware, and component-aware GPs without changing a design or observation.

C Deferred dynamic actions

The current environment records time-stamped interim measurements but exposes no treatment change, early termination, cleanup, residual-value, or asynchronous replacement action. Adding one requires explicit transition dynamics and a matched oracle. Until then, claims about the value of within-cycle observation are restricted to recommendation and next-round behaviour.